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QUANTUM ASSISTED REPRESENTATION THEORY OF E_8

Variational Quantum Eigensolvers for Spectral Estimation of E_8 Tensor-Product Casimir Operators

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Abstract

The decomposition of tensor products of the exceptional Lie algebra E_8 is a computational challenge in representation theory and mathematical physics, with potentially substantial memory and runtime demands for high-dimensional modules. Complexity depends on the algorithm, input encoding and required output; a universal exponential lower bound for classical approaches is not established here. This manuscript presents a deductive framework that formulates aspects of tensor product decomposition as a quantum spectral estimation problem. By mapping the tensor-product quadratic Casimir, constructed from coproduct generators, onto a multi-qubit system and using a Variational Quantum Eigensolver (VQE), the framework seeks to estimate spectral data relevant to constituent irreducible representations. A single Casimir spectrum need not distinguish all irreducible labels, and spectral projector traces give dimension-weighted sums of multiplicities. Recovering individual multiplicities therefore requires additional representation-theoretic information; polynomial-time full decomposition and global VQE convergence are not established. The framework connects abstract representation theory with fault-tolerant quantum information science through explicit generator-dependent Lipschitz bounds for optimization smoothness, ideal syndrome-extraction identities, and conditional error-correction resource models. Reproducible Python diagnostics support the finite mathematical checks, while complete fault-tolerant syndrome cycles and hardware performance guarantees remain implementation requirements.

To facilitate reproducibility, Python validation scripts accompany the mathematical results identified in Quantum Asissted Representation Theory of E_8 .The scripts implement the described algorithms and perform exact finite computations and numerical consistency checks supporting the analysis. These computational checks complement, but do not replace, the mathematical proofs and their stated assumptions. The code is available at:

https://github.com/teknoquants/Quantum-Assisted-Representation-Theory-of-E_8

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Introduction

The structural classification and decomposition of tensor products within exceptional Lie algebras connect representation theory, mathematical physics and quantum information. The E_8 root system has 240 roots spanning an eight-dimensional Euclidean space, and its adjoint representation has dimension 248. For finite-dimensional complex irreducible modules $V(\lambda)$ and $V(\mu)$, Weyl complete reducibility gives a direct sum of irreducibles with uniquely determined isomorphism classes and multiplicities, although the embedded summands need not be canonical.

Evaluating these multiplicities can require substantial classical computation for large modules. The E_8 Weyl group has order 696729600. This large but fixed constant does not itself prove an exponential lower bound for classical algorithms as highest weights grow. Complexity comparisons must identify an algorithm, input encoding, varying parameters and required output.

This manuscript studies a spectral framework in which the tensor-product Casimir is assembled from coproduct generators and represented on a quantum register. A variational eigensolver may provide spectral information, but the spectrum of one Casimir does not generally identify all irreducible labels or multiplicities. A spectral projector trace counts a dimension-weighted sum of multiplicities. Neither polynomial-time full decomposition nor global VQE convergence is established here.

A single 248-dimensional adjoint module admits an eight-qubit amplitude encoding. The adjoint tensor square has dimension 61504 and requires at least sixteen qubits in the same direct encoding. Excitation-conserving Givens blocks preserve Hamming-weight sectors, which are not identical to the first-248-state physical subspace. The ansatz and initialization therefore require explicit invariant-subspace checks and cannot rely on weight conservation alone to prevent padding leakage.

The fault-tolerant discussion uses a conditional distance-three rotated surface-code resource model: eight patches with nine data qubits and eight syndrome ancillas each require 136 qubits before an assumed eight additional ancillas. This 144-qubit budget is not a complete implementation proof. Generator-dependent smoothness bounds, ideal distillation models and decoding graph estimates are stated with their hypotheses; real-time latency, gate synthesis, routing and circuit-level noise thresholds require further specification.

The boundary-field-theory discussion separates Chern—Simons gauge-level matching from gravitational and thermal-response data. Flat holonomy need not be scalar, and scalar phases do not establish non-Abelian statistics. Conventional E_8 level-one theory has only the vacuum integrable sector. These mathematical observations do not construct a protected E_8 interface on the proposed hardware. The combined paper therefore provides a conditional framework and reproducible checks for exploring the spectral approach, rather than a demonstrated speedup or an experimentally certified processor.

Part I Corrected original framework

Framework scope

Validation ID GLOBAL | check_global.py

Justification. For representations ρ_λ and ρ_μ, the tensor generator is $\Delta(X) = \rho_\lambda(X) \otimes I + I \otimes \rho_\mu(X)$. The tensor Casimir must be built from these generators with a consistent invariant metric. Its eigenvalues partition the representation into spectral subspaces, but additional information is generally needed to recover irreducible labels and multiplicities. An adjoint-only 8-qubit calculation is not a calculation on the adjoint tensor square.

Theorem 1 Complete reducibility of tensor modules

Validation ID T1 | check_t1.py

Let g be the complex simple Lie algebra E_8 and $V(\lambda), V(\mu)$ finite dimensional irreducible highest weight modules. Their tensor product is isomorphic to a finite direct sum of $V(\nu)$ with uniquely determined nonnegative integer multiplicities $N(\lambda, \mu; \nu)$. Uniqueness concerns the isomorphism classes and multiplicities, not a canonical choice of embedded summands.

Justification. Weyl complete reducibility applies because g is semisimple and the tensor product is finite dimensional over a characteristic zero field. It gives existence of a direct sum decomposition. Schur lemma implies that the multiplicity of $V(\nu)$ is $\dim \mathrm{Hom}_g\big(V(\nu), V(\lambda) \otimes V(\mu)\big)$, so it is independent of the chosen decomposition. The scripts check a dimension consistency example only; they do not formalize Weyl theorem.

Lemma 1 Weyl group order and complexity qualification

Validation ID L1 | check_l1.py

The Weyl group of E_8 has order $2^{14} \cdot 3^5 \cdot 5^2 \cdot 7 = 696729600$. This constant alone gives no exponential lower bound for classical tensor decomposition algorithms. Complexity statements must specify the input encoding, algorithm, output and varying parameters.

Justification. The product of the E_8 invariant degrees 2, 8, 12, 14, 18, 20, 24, 30 gives the same integer. For fixed E_8 and a fixed dominant weight λ, the Weyl dimension formula for $V(t\lambda)$ is a product of 120 affine factors in t divided by constants. It is therefore polynomial in t, although not necessarily polynomial in the bit length of t. No claim that all algorithms enumerate the entire Weyl group is made.

Definition 1 Isometric register encoding

Validation ID D1 | check_d1.py

For a module of dimension d, choose $n = \lceil \log_2 d \rceil$ qubits and an orthonormal basis. Define $\Phi(e_j) = |j\rangle$ for $0 \leq j < d$, with zero padding on the remaining register states. For $d = 1$ take $n = 0$. A single adjoint E_8 module has $d = 248$ and fits in eight qubits. The adjoint tensor square has $d = 61504$ and requires at least sixteen qubits under this encoding.

Justification. The computational basis vectors are orthonormal, hence Φ preserves all inner products by linearity. Conversely, an isometry is injective and requires $2^n \geq d$. The dimension 248 refers to the adjoint module, not to every fundamental highest weight representation of E_8.

Theorem 2 Matrix unit Pauli expansion

Validation ID T2 | check_t2.py

Use most significant bit first tensor order. For n −bit indices i and j, $|i\rangle\langle j|$ is the tensor product of the one-qubit matrix units $|b_k(i)\rangle\langle b_k(j)|$. The local identities are $|0\rangle\langle 0| = \frac{I+Z}{2}$, $|1\rangle\langle 1| = \frac{I-Z}{2}$, $|1\rangle\langle 0| = \frac{X-iY}{2}$ and $|0\rangle\langle 1| = \frac{X+iY}{2}$.

Justification. Each local identity follows by direct multiplication of the 2 by 2 Pauli matrices. Taking tensor products yields a matrix with its only nonzero entry equal to one at row i and column j. This is binary matrix encoding; no fermionic Jordan−Wigner transformation is asserted. The scripts exhaustively check the eight-qubit index mapping.

Corollary 1 Spectral normalization

Validation ID C1 | check_c1.py

Let C be a nonzero positive semidefinite Hermitian Casimir matrix in a specified compact Hermitian generator convention. Set $c_{\max} = \|C\|$ and $\tilde{C} = C/c_{\max}$. Its spectrum lies in [0,1] and its maximum is one. The minimum need not be zero. For $C = 0$, define $\tilde{C} = 0$ separately.

Justification. Hermitian diagonalization gives eigenvalues $0 \leq c_j \leq c_{\max}$. Division gives $c_j/c_{\max}$ in [0,1]. The scale is fixed by requiring the maximum to be one, not merely by containment in $[-1,1]$. For an invariant metric in a nonorthonormal basis, the Casimir uses dual generators rather than an unqualified sum of squares.

Definition 2 Deflated objective

Validation ID D2 | check_d2.py

Given a specified padded Hermitian matrix H, normalized states ψ_j, real penalties β_j and the orthogonal projector Q onto padding states, define $H_{\text{def}} = H + \sum_j \beta_j |\psi_j\rangle\langle\psi_j| + \beta_{\text{leak}} Q$. Exact eigenvector deflation and approximate-state deflation are distinct settings.

Justification. Every outer product and Q is Hermitian, so H_{def} is Hermitian. If the ψ_j are orthonormal, their rank one projectors are pairwise orthogonal. The definition by itself imposes no convergence, no exact leakage suppression and no recovery of an entire irreducible summand.

Theorem 3 Exact vector deflation

Validation ID T3 | check_t3.py

Let H have orthonormal eigenvectors ψ_j and spectrum in $[a, b]$. Suppose a proper subset of these eigenvectors is penalized and at least one is unpenalized. If $\beta_j > b - \lambda_j$ for every penalized eigenvector, every global ground state of $H + \sum_j \beta_j |\psi_j\rangle\langle\psi_j|$ is orthogonal to their span. The common bound $\beta_j > b - a$ is sufficient.

Justification. The operator is diagonal in the eigenbasis of H. Penalized eigenvalues become $\lambda_j + \beta_j > b$; every unpenalized eigenvalue remains at most b. Thus none of the penalized vectors belongs to its ground space. The conclusion excludes only the penalized vectors, not other vectors in the same irrep or degenerate eigenspace. It does not guarantee that VQE finds a global minimum. On [0,1], $\beta = 2.5$ is sufficient but unnecessarily large; approximate eigenvectors need separate bounds.

Corollary 2 Conditional padding isolation

Validation ID C2 | check_c2.py

Assume H is block diagonal on physical and padding subspaces. Let a_{phys} be the minimum physical eigenvalue of H including any physical deflation terms, and a_{pad} the minimum padding eigenvalue before the padding penalty. If $a_{\text{pad}} + \beta_{\text{leak}} > a_{\text{phys}}$, every global ground state has zero padding support.

Justification. The two spectra remain separate blocks and the penalty shifts only the padding block. A strict comparison of their minimum eigenvalues proves the statement. A value $\beta_{\text{leak}} = 5$ works for zero padding and an available unpenalized physical eigenvalue at most one. Coupling the blocks can leave nonzero leakage at any finite penalty; restricted ansatz expressivity and optimization errors also prevent a general VQE guarantee.

Definition 3 Excitation conserving ansatz

Validation ID D3 | check_d3.py

Define $N = \sum_k \frac{I - Z_k}{2}$. A gate conserves Hamming weight when $[N, G] = 0$. This condition is distinct from $[\tilde{C}, G] = 0$. Casimir commutation is not imposed on an ansatz intended to change the Casimir expectation.

Justification. If U commutes with $\tilde{C}$, then $\langle\psi|U^\dagger \tilde{C} U|\psi\rangle = \langle\psi|\tilde{C}|\psi\rangle$. Such an ansatz cannot vary that objective from a fixed input. A circuit made only of excitation conserving gates also keeps the vacuum in the zero-excitation sector, so a nontrivial starting sector requires a preparation operation.

Theorem 4 Spectral projector dimension

Validation ID T4 | check_t4.py

If P_c projects onto a Casimir eigenspace, then $\mathrm{Tr}(P_c) = \sum_{\nu:\, c_\nu = c} N(\lambda, \mu; \nu) \dim V(\nu)$. For an orthonormal basis $\psi_1, \ldots, \psi_m$ of that eigenspace, $P_c = \sum_k |\psi_k\rangle\langle\psi_k|$ and $\mathrm{Tr}(P_c) = m$.

Justification. On each irreducible summand the central Casimir is scalar by Schur lemma. Every copy contributes its full module dimension to the corresponding eigenspace. Trace of each normalized rank one projector is one. If a single irrep label is known to occur at c, its multiplicity is $\mathrm{Tr}(P_c)/\dim V(\nu)$. Counting only highest weight vectors requires a separately specified highest weight restriction. Multiple runs alone do not certify completeness or orthogonality.

Corollary 3 Limits of subalgebra splitting

Validation ID C3 | check_c3.py

An $\mathfrak{so}(16)$ Casimir may refine some E_8 Casimir eigenspaces, but it is not guaranteed to distinguish all E_8 labels or repeated copies. The perturbation $C_{E_8} + \delta C_{\mathfrak{so}(16)}$ requires explicit branching data and a chosen normalization; no universal multiplicity formula follows from the number of split lines.

Justification. The E_8 Casimir commutes with represented subalgebra operators. On an isotypic component $V(\nu) \otimes \mathbb{C}^m$, every represented enveloping-algebra operator acts as an operator on $V(\nu) \otimes I_m$. Thus identical copies remain indistinguishable to this perturbation. Distinct branching components can also share subalgebra Casimir values.

Lemma 2 Explicit E_8 roots

Validation ID AL2 | check_al2.py

Use R_{int} consisting of all vectors with exactly two entries equal to plus or minus one and six zero entries. Use R_{half} consisting of all eight-tuples of plus or minus one half with an even number of negative signs. Their union has 240 vectors of squared length two and spans $\mathbb{R}^8$.

Justification. There are $\binom{8}{2} \cdot 4 = 112$ integer roots and $2^7 = 128$ half roots. The norm calculation gives two in both cases. In doubled integer coordinates the scripts also exhaustively verify crystallographic integrality and reflection closure for all 57600 ordered root pairs, as well as rank eight. These checks supplement the count and norm argument; equal lengths alone would not establish a root system.

Theorem 5 Excitation conserving Givens block

Validation ID T5 | check_t5.py

Define $G(\theta, \varphi)$ as the identity on $|00\rangle$ and $|11\rangle$, with block $\begin{bmatrix} \cos\theta & -e^{-i\phi}\sin\theta \\ e^{i\phi}\sin\theta & \cos\theta \end{bmatrix}$ on the ordered basis $|01\rangle, |10\rangle$. Then G is unitary and commutes with N. In general this complex family is not a real $\mathrm{SO}(4)$ matrix.

Justification. The columns of the central block are orthonormal. Both vectors in that block have excitation number one, while the other basis vectors remain fixed; hence the commutator with N is zero. The previously stated two-CNOT factorization is withdrawn: its literal tensor dimensions disagree, and removing the extra identity still yields excitation-changing transitions. No exact native gate count is asserted for this corrected family.

Lemma 7 Generator form of the Givens block

Validation ID AL7 | check_al7.py

Let K_φ be zero on $|00\rangle$ and $|11\rangle$ and equal to $\begin{bmatrix} 0 & -ie^{-i\phi} \\ ie^{i\phi} & 0 \end{bmatrix}$ on $|01\rangle, |10\rangle$. Then $G(\theta, \varphi) = \exp(-i\theta K_\varphi)$, with G as in Theorem 5.

Justification. K_φ is Hermitian and its square is the identity on the excitation-one block. The exponential there is $\cos(\theta)I - i\sin(\theta)K_\varphi$, which gives the stated block. On the other two states the exponential is the identity. This is a generator identity, not a hardware synthesis or a two-CNOT resource theorem.

Theorem 12 Necessary support rule for root generators

Validation ID T12 | check_t12.py

In a weight basis, a matrix element of E_α from weight ω to weight η can be nonzero only if $\eta = \omega + \alpha$. Root coordinates alone do not determine all coefficients, cocycle signs or actions on the eight-dimensional zero-weight Cartan space of the adjoint module.

Justification. For a weight vector v, the relation $[H, E_\alpha] = \alpha(H)E_\alpha$ implies $H(E_\alpha v) = \big(\omega(H) + \alpha(H)\big)E_\alpha v$. Thus a nonzero image has weight $\omega + \alpha$. The converse is false: in the adjoint representation $[E_\alpha, H] = -\alpha(H)E_\alpha$ can vanish for a nonzero H even though the weight difference is α. A complete construction requires a basis and compatible bracket constants.

Definition 4 Root metric Casimir convention

Validation ID D4 | check_d4.py

With long roots of squared length two, define $c_{\mathrm{root}}(\lambda) = (\lambda, \lambda + 2\rho)$. In this convention the adjoint value for E_8 is 60. If C_A is defined as one half of this Casimir, its adjoint value is 30, the dual Coxeter number.

Justification. The normalization factor must be part of the definition. The root enumeration computes ρ as half the positive-root sum and evaluates the highest-root expression as 60. The half-scaled convention then gives 30. These conventions must not be mixed with the trace-normalized basis below.

Definition 5 Adjoint trace normalization

Validation ID D5 | check_d5.py

Let X_a be an invariant orthonormal Hermitian generator basis acting on the irreducible 248-dimensional adjoint module, with $\mathrm{Tr}(X_a X_b) = \delta_{ab}$. Then $\sum_a X_a^2 = I_{248}$.

Justification. The invariant quadratic sum is central, hence a scalar cI by Schur lemma. Its trace is $\sum_a \mathrm{Tr}\,(X_a^2) = 248$, while $\mathrm{Tr}(cI) = 248c$, so $c = 1$. This argument assumes the specified basis and irreducibility; it does not construct the generators. A root basis E_α is paired with $E_{-\alpha}$ rather than being a Hermitian orthonormal basis itself.

Table 1 Generator dimensions and encoding

Validation ID TABLE1 | check_table1.py

The adjoint module has eight Cartan directions and 240 root directions, with the roots divided into 112 integer and 128 half-integer coordinates. Eight computational states are padding for an eight-qubit adjoint register. Cartan matrices are diagonal in a weight basis and have expansions in I and Z strings.

Justification. No fixed Pauli locality or dense-block property follows from integer versus half-integer root coordinates. Root generators obey weight-shift support, while their actual Pauli expansions depend on the matrices and basis order. Padding penalties are diagonal and also admit I/Z expansions. The dimensions add to 248, and $256 - 248 = 8$.

Proposition 1 Conditional surface code resource count

Validation ID P1 | check_p1.py

For eight independent distance-three rotated patches using nine data qubits and eight syndrome ancillas per patch, the baseline count is $8 \cdot (9 + 8) = 136$. Adding an assumed eight routing or flag ancillas gives 144. This is a resource budget under stated assumptions, not a proof that 144 qubits suffice for all logical gates.

Justification. The count is direct arithmetic. The sixteen logical qubits needed for a direct adjoint tensor-square encoding would require a $16 \cdot 17 = 272$ baseline under the same patch model, before routing and resource states. Distillation, gate synthesis, scheduling, connectivity and repeated syndrome extraction require separate resource analysis.

Theorem 6 Parameter dependent smoothness bound

Validation ID T6 | check_t6.py

For a normalized state ψ, Hermitian H and G, let $E(\theta) = \langle\psi|\exp(i\theta G)H\exp(-i\theta G)|\psi\rangle$. Then $|E''(\theta)| \leq 4 \parallel G \parallel^2 \parallel H \parallel$, so the derivative is globally Lipschitz with this bound. Parameter generator norms cannot be omitted.

Justification. Twice differentiating gives an expectation of a conjugated negative double commutator $[G, [G, H]]$. Using $\|[A, B]\| \leq 2\|A\|\|B\|$ proves the bound. For a fixed ordered product of one-parameter gates $\exp(-i\theta_j G_j)$, mixed derivatives are bounded by $4\|H\|\|G_j\|\|G_k\|$, giving the sufficient Euclidean gradient bound $L \leq 4 \parallel H \parallel \sum_j \parallel G_j \parallel^2$. For normalized deflation projectors, $\parallel H_{\text{def}} \parallel \leq \parallel H \parallel + \sum_j |\beta_j| + |\beta_{\text{leak}}|$. Smoothness is not a global convergence guarantee.

Lemma 3 Ideal syndrome extraction

Validation ID AL3 | check_al3.py

A Z parity check can be extracted by initializing an ancilla in $|0\rangle$, applying data-controlled CNOTs into it and measuring Z. An X parity check uses an ancilla in $|+\rangle$, ancilla-controlled CNOTs and an X −basis measurement, equivalently a final Hadamard followed by Z measurement.

Justification. For computational data bits the target ancilla accumulates their XOR, which is the eigenvalue sign of the Z product. Conjugating by Hadamards yields the X version. These identities describe ideal checks only. Boundary checks may have weight two; scheduling, hook errors, measurement errors, repeated rounds and timing must be addressed before claiming fault tolerance. There are 64 syndrome ancillas in the assumed eight-patch budget.

Theorem 7 Minimum weight matching objective

Validation ID T7 | check_t7.py

For an even defect set with a feasible matching graph, minimum weight perfect matching minimizes the sum of its supplied edge weights. Minimizers need not be unique. Under an independent edge error model with $0 < p_e < 1/2$, a chain cost uses $\log\big((1-p_e)/p_e\big)$, up to constants.

Justification. The probability of a chain with selected error edges is proportional to the product of $p_e/(1-p_e)$; its negative logarithm is additive. Translating to pairwise shortest paths additionally needs compatible boundary treatment. A square with unit side lengths has two optimal pairings of cost two, disproving uniqueness. Maximum likelihood of a chain differs from maximum likelihood of a logical class, which sums many chains. Matching cannot guarantee logical correctness for every syndrome and error.

Theorem 8 Conditional circuit scheduling count

Validation ID T8 | check_t8.py

One nearest-neighbor brickwork layer on eight qubits contains four disjoint even-pair blocks and three disjoint odd-pair blocks. If each intended block has a synthesis of g two-qubit gates and two-qubit depth t, D layers admit a schedule with at most $7gD$ gates and depth at most $2tD$, under parallel disjoint-pair execution.

Justification. Each parity block executes its four or three pair circuits concurrently. The two parity blocks are sequential, giving the stated count and depth. Values $g = t = 2$ would give $14D$ gates and $4D$ depth only after a valid synthesis is supplied. The withdrawn factorization does not supply one. Fault tolerant implementation and connectivity overhead may change these logical-level costs.

Lemma 5 Invariant root trace pairing

Validation ID L5 | check_l5.py

For an invariant trace form $B(X,Y) = \mathrm{Tr}\big(\rho(X)\rho(Y)\big)$, root vectors have $B\big(E_\alpha, E_\beta\big) = 0$ unless $\beta = -\alpha$. For nonzero opposite-root pairing, vectors may be scaled to satisfy $B(E_\alpha, E_{-\alpha}) = 1$; then $\mathrm{Tr}([H, E_\alpha]E_{-\alpha}) = \alpha(H)$. Bracket constants must use the same scaling.

Justification. Invariance gives $\big(\alpha(H) + \beta(H)\big)B\big(E_\alpha, E_\beta\big) = 0$ for every H. If $\alpha + \beta$ is nonzero, choose H where it does not vanish. For $\beta = -\alpha$ the last identity follows immediately from the root commutator relation and normalization. Bilinear trace orthogonality is not a statement that the associated Pauli coefficient supports are disjoint.

Theorem 9 Correct structure constant derivative

Validation ID T9 | check_t9.py

Assume $E_{-\alpha} = E_\alpha^\dagger$, the root trace pairing of Lemma 5, and $\alpha + \beta$ is a root. Put $K = E_\alpha + E_{-\alpha}, U(\theta) = \exp(-i\theta K)$, and $f(\theta) = \mathrm{Tr}\big(U(\theta)E_\beta U(\theta)^\dagger E_{-(\alpha+\beta)}\big)$. Then $f'(0) = -iN_{\alpha,\beta}$, so $N_{\alpha,\beta} = if'(0)$.

Justification. Differentiating at zero gives $-i\big[K, E_\beta\big]$. The E_α bracket contributes $-iN_{\alpha,\beta}$ by the normalized trace pairing; the $E_{-\alpha}$ bracket has a different weight and pairs to zero. No universal two-point parameter-shift formula is asserted. Such a rule requires generator spectral gaps and suitable shift coefficients; this trace can be complex.

Definition 6 CSS stabilizer checks

Validation ID AD6 | check_ad6.py

For data-qubit supports A and B, define $X(A)$ and $Z(B)$ as products of X and Z on their respective supports. They commute exactly when $|A \cap B|$ is even. A CSS code uses binary check matrices satisfying $H_X H_Z^T = 0$ over $\mathbb{F}_2$, together with independent compatible stabilizer generators.

Justification. Moving $X(A)$ past $Z(B)$ produces the sign (-1) raised to the overlap size. Thus even overlap is necessary and sufficient. The scripts specify an explicit nine-data-qubit patch with four independent X checks and four independent Z checks, verify all overlaps and exhaustively find CSS distances three. Boundary checks include weight two. The 144-qubit budget includes measurement ancillas, which are not all persistent encoded data qubits.

Proposition 10 Conditions on geometric holonomy

Validation ID T10 | check_t10.py

A connection defines a path-ordered holonomy around a closed loop. Flatness on the allowed region gives homotopy invariance for based loops under the usual connection assumptions, but does not imply scalar holonomy. A scalar phase requires an additional restriction on the holonomy representation. No E_8 hardware protection follows without an explicit connection and physical model.

Justification. On an annulus, take $A = \frac{iZ\, d\phi}{8}$. The curvature vanishes, yet the closed-loop holonomy is $\mathrm{diag}\left(e^{i\pi/4}, e^{-i\pi/4}\right)$, which is not scalar. For an adiabatic implementation, spectral gaps, path control and errors require their own analysis. The earlier scalar and noise-protection assertions are therefore withdrawn.

Requirement 6 State preparation

Validation ID AL6 | check_al6.py

A preparation unitary U_{init} must be explicitly specified and checked against a target normalized vector ψ_{target} in the encoded physical subspace. Exact preparation requires $U_{\mathrm{init}}|0 \cdots 0\rangle = \psi_{\mathrm{target}}$ up to global phase; padding avoidance alone requires $QU_{\mathrm{init}}|0 \cdots 0\rangle = 0$.

Justification. The vacuum has no padding support in the chosen ordering, but that does not identify it with a desired highest weight vector. Gates commuting with excitation number cannot move it into a nonzero-excitation sector. Root intersection angles do not define a unique preparation circuit. Initialization remains an implementation requirement until a target and circuit are provided.

Requirement 11 Logical error scaling and threshold

Validation ID T11 | check_t11.py

A distance-three code with an appropriate fault tolerant extraction circuit and decoder may have leading logical failure probability of order p^2 per specified round or experiment. The coefficient and threshold depend on the full noise, circuit and decoder model. No numerical 1.1 percent threshold is established here.

Justification. If all single faults are correctable and some two-fault events fail, a small$-p$ expansion begins quadratically. Distance of the data code alone does not establish circuit-level fault tolerance. Threshold estimation requires a family of distances and explicit experiments with uncertainty estimates. Scaling in physical error p must not be confused with scaling in runtime or circuit depth. The scripts illustrate the quadratic expansion with an exactly enumerated repetition-code example, not a surface-code threshold simulation.

Proposition 2 Ideal phase error distillation model

Validation ID P2 | check_p2.py

Consider a specified ideal 15-to-1 phase-error model with independent input errors of probability p and perfect Clifford operations. Label the 15 positions by the nonzero four-bit vectors. Accept error patterns whose label XOR is zero; an accepted odd-weight pattern is a logical error. Let A_w and B_w count accepted and accepted odd patterns of weight w. Then $p_{\text{out}} = \frac{\sum_w B_w p^w (1-p)^{15-w}}{\sum_w A_w p^w (1-p)^{15-w}}$.

Justification. Enumeration gives $B_1 = B_2 = 0$ and $B_3 = 35$, with $A_0 = 1$, so $p_{\text{out}} = 35p^3 + O(p^4)$. The nontrivial fixed point of the full ratio is approximately 0.14148. The truncated equation $p = 35p^3$ instead gives $1/\sqrt{35}$, approximately 0.16903, and cannot justify 14.1 percent. A conventional T resource is $T|+\rangle = \frac{|0\rangle + e^{i\pi/4}|1\rangle}{\sqrt{2}}$; the real state $\cos(\pi/8)|0\rangle + \sin(\pi/8)|1\rangle$ uses an H −type convention and needs an explicit conversion/noise convention. This model establishes neither mandatory use of this protocol nor a complete resource budget for arbitrary-angle rotations.

Lemma 4 Correct Cartan diagonal and Pauli expansion

Validation ID L4 | check_l4.py

Order the 240 roots lexicographically, followed by eight zero Cartan weights and eight padding entries. Let $h_j^{(a)}$ be the a-th root coordinate divided by $\sqrt{60}$, with zero entries for the last sixteen states. Define $H_a = \sum_j h_j^{(a)} |j\rangle\langle j|$. Then $\mathrm{Tr}(H_a H_b) = \delta_{ab}$. For $n = 8$, $H_a = \sum_s c_s^{(a)} Z^s$, where $c_s^{(a)} = 2^{-8} \sum_j h_j^{(a)} (-1)^{\mathrm{popcount}(s \text{ AND } j)}$.

Justification. The exact root coordinate Gram matrix is $60I$, giving the trace normalization. Each projector is the tensor product of $\frac{I + (-1)^b Z}{2}$; the previously stated expression instead selected I or Z and was not a projector. The binary characters obey $\sum_j \chi_s(j)\chi_t(j) = 256\delta_{st}$, proving the Walsh coefficient formula and reconstruction. The scripts verify the Gram matrix and a representative reconstructed Cartan generator. This ordering is an explicit convention, not an inferred hardware layout.

Part II Corrected updated results

The eight supporting results below restate dependencies used by the update. They are retained to preserve the updated proofs and their validation mapping, and do not replace the corresponding fuller treatments in Part I.

Supporting result 1 E_8 algebraic data and reducibility

Validation F1 | check_f1.py

Work over the complex numbers. The adjoint E_8 module has dimension 248 and rank 8. Its root system consists of 112 integer roots with two entries ± 1 and 128 half-integer roots with all entries $\pm 1/2$ and an even number of negative signs. Every root has squared length two. Finite dimensional E_8 modules are completely reducible, with unique isomorphism types and multiplicities, although their embedded summands need not be canonical.

$$248 = 8 + 112 + 128$$

$$|W(E_8)| = 2^{14} \times 3^5 \times 5^2 \times 7 = 696729600$$

$$V(\lambda) \otimes V(\mu) \cong \oplus_\nu V(\nu) \otimes \mathbb{C}^{N(\lambda,\mu;\nu)}$$

Proof. There are 4 times $\binom{8}{2}$ integer roots and 2^7 half-integer roots. Direct calculation gives squared length two. The finite root data can also be checked for reflection closure. Weyl complete reducibility gives the direct sum, and Schur lemma identifies each multiplicity with the dimension of its intertwiner space. The Weyl group order is the product of invariant degrees 2,8,12,14,18,20,24,30. A fixed Weyl group order is not a complexity lower bound as weights vary.

Scope. Complete reducibility is an invoked general theorem, not a conclusion of a finite Python test. No universal exponential classical lower bound is established.

Computational check. Exact root counts, lengths, Gram matrix and all 57600 root reflections; integer Weyl order.

Supporting result 2 Encoding and excitation sectors

Validation F2 | check_f2.py

A d −dimensional module can be encoded isometrically in $n = \lceil \log_2 d \rceil$ qubits by sending an orthonormal basis to the first d computational vectors. Eight qubits accommodate one adjoint module; the adjoint tensor square requires sixteen. Hamming-weight conservation preserves sectors of dimension $\binom{n}{w}$, not the arbitrary first-248-state subspace.

$$248^2 = 61504 > 2^8; \quad 2^{15} < 61504 \leq 2^{16}$$

$$\dim H_w = \binom{n}{w}; \quad \max_w \binom{8}{w} = 70$$

$$N = \sum_j \frac{I - Z_j}{2}$$

Proof. An isometry is injective, so the target dimension must be at least d. Orthonormal basis encoding realizes this bound. A weight−w basis state is specified by choosing w excited positions. For a concrete leakage counterexample, states |247⟩ and |251⟩ have weight seven; swapping their adjacent differing bits is excitation conserving but takes a valid index into padding. Thus $[U, N] = 0$ does not imply invariance of the physical projector.

Scope. The 248-dimensional representation is adjoint. It is not the dimension of every fundamental E_8 representation. A physical-subspace invariant ansatz must be independently established.

Computational check. Dimension arithmetic, every sector size and an explicit 247-to-251 excitation-conserving transition.

Supporting result 3 Casimir normalization and multiplicities

Validation F3 | check_f3.py

For a compact Hermitian generator convention with its specified invariant metric, the Casimir is central. On a tensor product use the coproduct generators. A nonzero positive Casimir C can be normalized by its largest eigenvalue to spectrum in [0,1]. The trace of a Casimir spectral projector counts full eigenspace dimension, not generally irreducible multiplicity.

$$\Delta(X) = \rho_\lambda(X) \otimes I + I \otimes \rho_\mu(X)$$

$$\mathrm{Tr}(P_c) = \sum_{\nu:\, c_\nu = c} N(\lambda, \mu; \nu) \dim V(\nu)$$

$$c_{\mathrm{root}}(\lambda) = (\lambda, \lambda + 2\rho); \quad c_{\mathrm{root}}(\mathrm{adj}) = 60$$

$$C_A = \frac{C_{\mathrm{root}}}{2}; \quad c_A(\mathrm{adj}) = 30$$

Proof. Centrality and Schur lemma give a scalar on each irreducible copy, so each copy contributes its dimension to the projector trace. Division by the largest positive eigenvalue gives [0,1]; the zero operator is handled separately. Roots of squared length two give the root-metric adjoint value 60. Halving the invariant metric convention gives 30. If instead $\mathrm{Tr}(X_a X_b) = \delta_{ab}$ in the adjoint matrix representation, taking the trace of the central sum of 248 squares gives $C = I$. These normalizations must not be mixed.

Scope. The single-adjoint Casimir is scalar and has no nontrivial irrep decomposition to discover. Full decomposition requires more information than one Casimir spectrum; no polynomial-time VQE guarantee follows.

Computational check. Root-metric Casimir, scale arithmetic, projector trace counterexample and normalization examples.

Supporting result 4 Deflation and padding hypotheses

Validation F4 | check_f4.py

Let H have spectrum in $[a, b]$ and let a proper subset of exact orthonormal eigenvectors ψ_j with eigenvalues λ_j be penalized. Penalties $\beta_j > b - \lambda_j$ exclude their span from every global ground state. For block-diagonal physical and padding sectors, a padding penalty excludes padding only if its shifted minimum is strictly above an available physical minimum.

$$H_{\mathrm{def}} = H + \sum_j \beta_j \left|\psi_j\right\rangle\!\left\langle\psi_j\right| + \beta_{\mathrm{leak}} Q$$

$$\beta_j > b - \lambda_j$$

$$a_{\mathrm{pad}} + \beta_{\mathrm{leak}} > a_{\mathrm{phys}}$$

Proof. In the exact eigenbasis, deflation changes λ_j to $\lambda_j + \beta_j > b$ and leaves at least one unpenalized eigenvalue at most b. For invariant physical/padding blocks, compare their two minimum eigenvalues after the shift. The value $\beta_{\mathrm{leak}} = 5$ is sufficient for zero padding when an accessible, unpenalized physical eigenvalue is at most one. Coupled blocks need not have exactly zero leakage at any finite penalty.

Scope. These are global spectral statements, not VQE convergence guarantees. One vector does not exhaust a degenerate eigenspace. Approximate eigenvectors and restricted ansatz spaces need separate bounds.

Computational check. Exact vector deflation and both invariant-block and coupled-block leakage examples.

Supporting result 5 Correct Givens gates and smoothness

Validation F5 | check_f5.py

Use a two-qubit unitary that is identity on $|00\rangle$ and $|11\rangle$, with the block below on $|01\rangle$, $|10\rangle$. It conserves excitation number. For a circuit with independent real parameters each entering one fixed Hermitian-generator exponential, the gradient Lipschitz constant depends on those generator norms. Casimir commutation would make its expectation constant, so it is not imposed on an optimizing ansatz.

$$G_{\mathrm{ex}}(\theta,\phi)=\begin{bmatrix}\cos\theta & -e^{-i\phi}\sin\theta\\ e^{i\phi}\sin\theta & \cos\theta\end{bmatrix}$$

$$|E''(\theta)|\le 4\,\|H\|\,\|G\|^2 \quad \text{for } U(\theta)=\exp(-i\theta G)$$

$$L\le 4\,\|H\|\sum_j \|G_j\|^2 \quad \text{for a fixed ordered product}$$

$$\text{gate count}\le 7gD;\quad \text{two-qubit depth}\le 2tD \quad \text{on 8 qubits}$$

Proof. The two central columns of G are orthonormal and have the same excitation number. Differentiating an expectation twice gives double commutators, bounded with $\|[A,B]\|\le 2\|A\|\|B\|$. Mixed derivatives are at most $4\|H\|\|G_j\|\|G_k\|$; the induced Hessian norm is bounded by the displayed rank-one majorant. Four disjoint even pairs and three disjoint odd pairs per layer give the conditional gate count when each block has g gates and depth t. A trace derivative under $U=\exp(-i\theta K)$ is $-i\mathrm{Tr}([K,B]D)$, not $\mathrm{Tr}([K,B]D)$; any two-point shift rule needs additional spectral assumptions.

Scope. The original two-CNOT synthesis is not used. The smoothness bound requires the specified parameterization, and does not prove global or exponential optimization convergence.

Computational check. Complex Givens unitarity and number commutation, generator-scaled Hessian examples, scheduling arithmetic and derivative factor.

Supporting result 6 Surface code checks and resource assumptions

Validation F6 | check_f6.py

For CSS stabilizers, $X(A)$ and $Z(B)$ commute if and only if their data-qubit support overlap is even. An assumed distance-three rotated patch with nine data qubits and eight syndrome ancillas has 17 qubits. Eight such patches need 136 qubits; adding eight unspecified auxiliary qubits yields a conditional budget of 144.

$$X(A)Z(B)=(-1)^{|A\cap B|}Z(B)X(A)$$

$$H_X H_Z^T=0 \quad \text{over } \mathbb{F}_2$$

$$8\times(9+8)=136;\quad 136+8=144$$

Proof. Every overlapping X/Z pair supplies one minus sign. The resource formulas then follow by counting. Ideal Z checks use a $|0\rangle$ target ancilla and data-controlled CNOTs; ideal X checks use a $|+\rangle$ control ancilla and X readout. A concrete nine-data-qubit CSS example is supplied in the scripts, with independent checks and exhaustively verified distance three.

Scope. The budget does not establish routing, gate synthesis, distillation or circuit-level fault tolerance. Spatial separation, code distance and decoding-window length are distinct variables. A physical-qubit count alone supplies no noise threshold.

Computational check. CSS parity, check ranks, logical distance and resource counts; ideal syndrome parity extraction.

Supporting result 7 Exact ideal distillation model

Validation F7 | check_f7.py

For a specified ideal 15-to-1 phase-error model, label the 15 positions by the nonzero four-bit vectors. Accept an error subset when the XOR of its labels is zero; an accepted odd-cardinality subset is a logical failure. Let A_w count accepted weight$-w$ subsets and B_w count accepted odd-weight subsets. Independent phase errors have probability p at every input.

$$a(p) = \sum_w A_w\, p^w (1-p)^{15-w}$$

$$f(p) = \frac{\sum_w B_w\, p^w (1-p)^{15-w}}{a(p)}$$

$$f(p) = 35p^3 + 105p^4 + O(p^5)$$

Proof. Enumeration gives $A_0 = 1, A_1 = A_2 = 0$ and $B_3 = 35$. The numerator starts as $35p^3 - 420p^4$, whereas $a(p) = 1 - 15p + O(p^2)$. Dividing yields the displayed series. A nonzero fixed point near 0.14 solves $f(p) = p$ and is approximately 0.14148. The cubic truncation alone instead has fixed point $1/\sqrt{35}$, approximately 0.16903. The positive p^4 correction prevents treating $35p^3$ as a strict upper bound near zero.

Scope. Perfect Clifford operations, independent twirled phase errors and postselection are assumptions. This model does not certify a physical distillation factory or arbitrary input noise.

Computational check. All 32768 input error patterns, exact series coefficients, rational probability evaluations and numerical fixed-point bracket.

Supporting result 8 Cartan normalization and diagonal coordinates

Validation F8 | check_f8.py

In the adjoint weight basis, order the 240 roots lexicographically, then eight zero Cartan weights, and finally eight padding states. Define diagonal H_a with root-coordinate entries divided by $\sqrt{60}$ and zeros on the last sixteen indices. These eight matrices have $\mathrm{Tr}(H_a H_b) = \delta_{ab}$ and commute, but do not span the full diagonal matrix algebra.

$$\sum_\alpha \alpha_a\, \alpha_b = 60\delta_{ab}$$

$$H_a = \sum_j h_j^{(a)}\, |j\rangle\langle j|$$

$$|j\rangle\langle j| = \otimes_k \frac{I + (-1)^{b_k(j)} Z}{2}$$

Proof. The root-coordinate Gram matrix is $60I$. Scaling the entries yields the trace pairing. The tensor product projector formula follows from the one-qubit identities. The eight zero weights are repeated

weights of the adjoint module, not eight distinct nonzero coordinates. Cartan matrices are linear combinations of Z strings, not generally single Pauli strings.

Scope. A Cartan subalgebra is maximal toral within the Lie algebra, not the center of all represented matrices. The Lie-algebra center of simple E_8 is zero.

Computational check. Exact coordinate Gram matrix, eightfold zero-weight multiplicity and rank of the Cartan diagonal space.

Updated theorems and proofs

Theorem 13 Separate level matching and WZW central charge

Validation T13 | check_t13.py

Assume a compact simply connected E_8 Chern−Simons theory with the basic invariant form that makes the level integral, together with its conventional affine WZW boundary construction. For a positive integer level k, the Sugawara central charge is $c(k) = 248k/(k + 30)$. Across an oriented interface the gauge anomaly coefficient is proportional to the oriented level difference Δk. This does not imply a relation $\Delta k \cdot 30 = c(k_{\text{inner}})(k_{\text{inner}} + 30)$.

$$c(k) = \frac{248k}{k + 30}; \quad c(1) = 8$$

$$\Delta k = k_{\text{inner}} - k_{\text{outer}}$$

$$\text{at } k_{\text{inner}} = 1 \text{ and } k_{\text{outer}} = 0: \quad 30 \neq 8 \times 31 = 248$$

Proof. In the standard affine current normalization, the Sugawara stress tensor is $T = \frac{1}{2(k+30)} \sum_a :J^a J^a:$.

The current algebra operator-product calculation gives central charge $\frac{k \dim(E_8)}{k+30}$. Substitution gives $c(1) = 8$. Chern−Simons gauge variations are linear in the action level; the two sides enter an interface with opposite orientations, so their coefficients subtract. Matching that coefficient requires a boundary anomaly of the opposite sign. Central charge controls a gravitational anomaly and is a separate datum. The explicit level-one calculation disproves the original equation.

Scope. The Sugawara/current-algebra construction is an invoked framework, not a derivation of a physical E_8 interface from root coordinates or qubits. An interface spectrum, boundary conditions, invariant form and orientation must be supplied; root-space intersections alone do not define a level jump.

Computational check. Exact rational $c(k)$, integer level differences and the level-one counterexample.

Reference basis: R1, R2.

Lemma 6 Decoder graph size and conditional complexity

Validation L6 | check_l6.py

Let s stabilizer outcomes be compared at each of R consecutive time steps. Before adding virtual boundary vertices, the detection-event count obeys $V_{\text{real}} \leq sR$. For the assumed eight-patch model $s = 64$. A complete graph on V vertices has $V(V-1)/2$ edges. If a selected weighted matching implementation has $O(V^3)$ arithmetic complexity, that bound applies to its matching stage; it is not an exact wall-clock latency.

$$V_{\text{real}} \leq 64R; \quad V = V_{\text{real}} + V_{\text{boundary}}$$

$$|E| = \frac{V(V-1)}{2}$$

$$T_{\text{total}} = T_{\text{graph}} + T_{\text{matching}} + T_{\text{other}}$$

Proof. Each stabilizer comparison contributes at most one detection event, so sR is a counting bound. Boundary nodes depend on the graph construction. The manuscript stated 64 ancillas but then multiplied 36 by two; neither that arithmetic nor a fixed $V \leq 72$ follows. Weighted blossom implementations also perform dual updates and contractions, so one bare breadth-first-search count is not a sufficient general complexity proof. For example, the documented NetworkX max-weight routine has $O(V^3)$ complexity; negating weights and enforcing maximum cardinality gives minimum-weight perfect matching when a perfect matching exists.

Scope. A finite graph size does not certify latency below a coherence time. Weight bit lengths, precision, graph construction, implementation and processor measurements are needed. A decoding window may contain many rounds.

Computational check. Adversarial syndrome histories saturating $64R$, graph counts, exhaustive small matching oracle and optional NetworkX comparison.

Reference basis: R3.

Theorem 10 Flat holonomy need not be scalar

Validation T10 | check_t10.py

For a connection on a specified bundle, flatness on an allowed region makes the holonomy representation depend on based-loop homotopy classes. It does not force the holonomy of a loop to be a scalar matrix. Scalarity requires a further restriction on that representation. Deformations crossing singularities or leaving the gapped adiabatic regime are not covered.

$$A = \frac{iZ\, d\phi}{8} \text{ on an annulus;} \quad F = dA + A \wedge A = 0$$

$$U(\gamma) = \text{diag}\left(e^{i\pi/4}, e^{-i\pi/4}\right) \neq e^{i\phi_0} I$$

Proof. On the annulus $d(d\varphi) = 0$ locally and $d\varphi \wedge d\varphi = 0$, so this connection is flat. Its matrices commute along the circular path, and the path-ordered exponential with the displayed positive-exponent convention reduces to $\exp(iZ \cdot 2\pi/8)$. Its two eigenvalues are unequal. This is a counterexample to the implication from flatness to scalar holonomy. General based-loop homotopy invariance follows from zero curvature; changing the base trivialization conjugates the holonomy.

Scope. No connection or adiabatic Hamiltonian is supplied for the hardware. Neither a scalar phase nor absolute immunity to control noise is established.

Computational check. Explicit unitary holonomy and its unequal eigenvalues; flat diagonal-loop integration.

Theorem 14 Conditional gradient concentration without convergence

Validation T14 | check_t14.py

Let E be smooth on the flat m −torus with period 2π in every coordinate, with gradient L −Lipschitz in the geodesic metric under its canonical tangent-frame identification. Let $\theta_1, \ldots, \theta_m$ be independent uniform angles and $f(\theta) = \|\nabla E(\theta)\|$. For $L > 0$ the following concentration bound holds. It is a statement about this parameter distribution, not Haar measure on an unspecified circuit group.

$$\Pr(|f - \mathbb{E}[f]| \geq \varepsilon) \leq \min\left(1, 2\exp\left(-\frac{2\varepsilon^2}{m\pi^2 L^2}\right)\right)$$

Proof. The reverse triangle inequality makes fL −Lipschitz. Altering only one torus coordinate changes geodesic distance by at most π, so the corresponding bounded difference is at most πL. Reveal the independent coordinates successively and form the Doob martingale. Each conditional martingale increment has range length at most πL. Hoeffding lemma and the exponential Markov bound give one-sided tail $\exp\left(-\frac{2\varepsilon^2}{\sum_j (\pi L)^2}\right)$; combine the two tails and cap by one. If $L = 0$, f is constant and every positive-deviation probability is zero. The Lie-algebra dimension 248 is not a substitute for m or a measure-concentration theorem on the actual ansatz manifold.

Scope. Concentration does not imply exponential descent or a global minimum. For $E(\theta) = \cos\theta$, initialization at $\theta = 0$ has zero gradient at a maximum. The cited barren-plateau work concerns small gradients under specific randomness assumptions and is not a convergence guarantee.

Computational check. Finite product-distribution concentration example, torus Lipschitz inequalities and a stationary maximum counterexample.

Reference basis: R4.

Proposition 15 Conditional thermal response and distinct anomalies

Validation T15 | check_t15.py

Gauge-current anomalies and gravitational anomalies are different Ward identities. Central charge alone does not specify a conserved electric charge, its current or an electrical Hall coefficient. If a physical interface realizes an equilibrated, ideal chiral edge with net central charge c_{minus}, the low-temperature thermal Hall response is the expression below. For a purely right-moving E_8 level-one edge, $c_{\text{minus}} = 8$.

$$\frac{\kappa_{xy}}{T} = \frac{\pi^2 k_B^2}{3h} c_{\text{minus}}$$

$$j_E(T) = \frac{\pi^2 k_B^2}{6h} c_{\text{minus}} T^2$$

Proof. For an ideal chiral boson channel, the energy-flux integral is $\frac{1}{h}\int_0^\infty \frac{E}{\exp(E/(k_B T))-1}\, dE$. Substituting $x = E/(k_B T)$ and expanding $\frac{1}{e^x - 1} = \sum_{n\geq 1} e^{-nx}$ gives $\sum_{n\geq 1} \frac{1}{n^2} = \frac{\pi^2}{6}$. Differentiating the flux with respect to T gives the thermal conductance per channel. In the stated chiral CFT setting the net channel coefficient is c_{minus}. Variation with respect to a metric defines stress-energy response, not a generic charge current. Hence the original equation equating divergence of an unspecified charge current to a spin-connection curvature is not established.

Scope. This is conditional edge physics. No E_8 edge, electromagnetic coupling, thermal equilibration or Hall experiment has been constructed on the proposed register. Orientation reverses the sign of c_{minus}.

Computational check. Bose-integral series enclosure and the factor of two relating energy flux to thermal conductance; no hardware-response simulation.

Reference basis: R5.

Theorem 16 Certified cubic distillation bound and resource growth

Validation T16 | check_t16.py

Let f be the exact conditional error map in Supporting result 7. Choose $0 < p_{\text{star}} < 1$ and $A > 0$ such that $f(p) \le Ap^3$ for $0 \le p \le p_{\text{star}}$ and $\sqrt{A}p_{\text{star}} < 1$. Starting at $p_0 \le p_{\text{star}}$, iteration gives the bound below. It concerns the number of distillation levels, not total gates, raw states or guaranteed elapsed time.

$$p_n \le A^{-1/2}\left(\sqrt{A}\,p_0\right)^{3^n}$$

$$A = \frac{\sum_{w\ge 3} B_w\, p_{\text{star}}^{w-3}}{(1-p_{\text{star}})^{15}}$$

$$n = O\left(\text{loglog}(1/\varepsilon)\right) \quad \text{for fixed } 0 < p_0 \le p_{\text{star}}$$

$$\text{raw inputs without rejection} = 15^n$$

$$\text{expected raw inputs} = \prod_{j=0}^{n-1} \frac{15}{a(p_j)}$$

Proof. The exact numerator is at most $p^3 \sum_{w\ge 3} B_w\, p_{\text{star}}^{w-3}$, because each factor $(1-p)^{15-w} \le 1$. Acceptance is at least $(1-p)^{15} \ge (1-p_{\text{star}})^{15}$. This proves the explicit A bound; $p_{\text{star}} = 0.01$ satisfies the contraction condition. With $x_n = \sqrt{A}\,p_n$, $x_{n+1} \le x_n^3$; induction proves the displayed inequality and keeps the iterates in range. Taking logarithms yields the level count. A depth$-n$ complete tree consumes 15^n raw states even before rejection; independent retries multiply expected consumption by 1/a at each level. Arbitrary ε means an error bound, not exact purity after finitely many rounds.

Scope. The coefficient 35 is only leading order and cannot replace A in a rigorous upper bound. Parallel depth may scale with levels only under explicit factory, scheduling and code-distance assumptions. Noise in Clifford operations can create a nonzero error floor.

Computational check. Exact Fraction upper bound, small$-p$ counterexample to $35p^3$, stable logarithmic recurrence, round estimate and expected resource calculation.

Theorem 17 Complete diagonal Pauli basis

Validation T17 | check_t17.py

All 256 strings in $\{I, Z\}^{\otimes 8}$ form an orthogonal basis for the 256-dimensional space of diagonal operators. Every diagonal matrix has the unique Walsh expansion below. The eight Cartan generators together with the single padding projector span at most nine dimensions, so they cannot represent all diagonal operators commuting with the adjoint Casimir.

$$D = \sum_{s=0}^{255} \hat{d}_s\, Z^s$$

$$\hat{d}_s = \frac{1}{256}\sum_{j=0}^{255} d_j\,(-1)^{\text{popcount}(s \text{ AND } j)}$$

$$\text{Tr}(Z^s Z^t) = 256\delta_{st}$$

Proof. For $s \ne t$, pairing binary strings that differ in a bit of s XOR t cancels their character sum; for $s = t$ every term is one. Orthogonality proves independence, uniqueness and reconstruction. In the adjoint encoding the Casimir is scalar on the physical block, so all physical diagonal matrices commute with it. The physical identity cannot lie in the Cartan span: all Cartan matrices have zero physical trace, while that

identity has trace 248. The padding diagonal algebra itself has dimension eight; its identity Q is only one direction. Repeated zero weights further invalidate the claim of 248 distinct Cartan weights.

Scope. The Lie-algebra center of E_8 is zero. Maximal toral inside E_8 does not mean maximal diagonal inside End($\mathbb{C}^{248}$). Diagonal Pauli completeness does not solve tensor-product multiplicities.

Computational check. Full 256-by-256 Walsh orthogonality, arbitrary diagonal reconstruction, Cartan-plus-padding rank and trace obstruction.

Theorem 18 Exponential path bound under explicit noise assumptions

Validation T18 | check_t18.py

Let s be a positive integer denoting physical separation, independently of code distance. Suppose every event of the specified connecting-error type includes a path of length $\ell \geq s$, the number of candidate paths of each length is at most $A_0\mu^{\ell}$, and the probability of any fixed path is at most p^{ℓ}. Assume $A_0 > 0$ and $0 < \mu p < 1$. Then the event probability obeys the bound below.

$$\Pr(\text{connection}) \leq \min\left(1, \frac{A_0(\mu p)^s}{1-\mu p}\right)$$

$$\xi = \frac{1}{\log\big(1/(\mu p)\big)}; \quad C = \frac{A_0}{1-\mu p}$$

Proof. Apply the union bound to all candidate paths and sum the geometric series from s to infinity. Writing $(\mu p)^s = \exp(-s/\xi)$ yields exponential separation dependence. The fixed-path probability assumption must be justified by an actual noise model. Ground-state exponential clustering concerns connected correlations of local observables under locality and gap hypotheses; it does not by itself bound driven, nonlocal logical cross-talk. For example, a shared logical-error channel acting on two patches with probability q independent of separation defeats every bound tending to zero with s.

Scope. The conditional path event need not equal all logical faults. Code distance three is not a declaration of separation three, a noise correlation length, or zero correlated logical error. Gap size alone does not specify a hardware cross-talk channel.

Computational check. Geometric-series bound and correlated-error counterexample at increasing separation.

Reference basis: R6.

Theorem 19 Scalar phases and level one integrability

Validation T19 | check_t19.py

A braid representation consisting solely of scalar matrices has commuting image and does not establish non-Abelian statistics. In the conventional E_8 affine theory at level one the vacuum is the only integrable highest-weight sector. The finite-dimensional adjoint is not an allowed independent level-one primary; the dimension-one currents occur as descendants in the vacuum sector.

$$\text{integrability: } \sum_i a_i^{\vee}\lambda_i \leq k, \quad \lambda_i \in \mathbb{Z}_{\geq 0}$$

E_8 comarks in a standard labeling: (2,3,4,6,5,4,3,2)

$$\text{at } k = 1: \quad \lambda_i = 0 \text{ for all } i$$

$$h_{\lambda} = \frac{c_{\text{root}}(\lambda)}{2(k+30)} = \frac{c_A(\lambda)}{k+30}$$

Proof. Every E_8 comark is at least two, so any nonzero Dynkin label violates the level-one integrability inequality. The adjoint highest weight is the highest root θ and has $(\theta, \theta^\vee) = 2$, requiring $k \geq 2$. The conformal-weight formula uses $60/\big(2(k+30)\big)$ or $30/(k+30)$ for an allowed adjoint primary, not $30/\big(2(k+30)\big)$; at level one that primary is absent regardless of arithmetic. Multiples of the identity commute with every matrix. Thus the proposed uniform $\exp(i\pi/15)$ phase cannot demonstrate a noncommuting braid action, and neither $30/62$ nor a root-lattice argument derives it.

Scope. The level-one bulk has no nontrivial anyon sector in this standard theory. Ordinary surface-code e and m excitations are Abelian; engineered defects would require a separately specified model. No mapping from an E_8 weight basis to anyonic fusion spaces is provided.

Computational check. Integer level-one integrability, adjoint-level condition, Casimir-convention arithmetic and scalar commutators.

Reference basis: R2.

Theorem 20 Conditional simulation and sampling costs

Validation T20 | check_t20.py

Specify an invariant N −dimensional state space, G elementary gates each applicable in $O(N)$ arithmetic work, an expectation-evaluation cost $C_H(N)$, and R decoding calls with cost $C_{\mathrm{dec}}(V)$. One state-vector trajectory has the cost below. If energy is estimated from independent unbiased outcomes in $[-B, B]$, with $B > 0, \varepsilon > 0$ and $0 < \delta < 1, S$ samples suffice for additive error ε and failure probability at most δ. A deterministic matrix expectation does not intrinsically require shot sampling.

$$T_{\mathrm{trajectory}} = O\big(GN + C_H(N) + RC_{\mathrm{dec}}(V)\big); \quad \text{state storage} = O(N)$$

$$S \geq \lceil \frac{2B^2}{\varepsilon^2} \log(2/\delta) \rceil$$

$$T_{\mathrm{sampled}} = O(S[GN + C_H(N) + RC_{\mathrm{dec}}(V)])$$

$$G = O(nD); \quad C_H(N) = O(N^2) \text{ for a dense matrix}$$

Proof. Add the costs of gate applications, one matrix expectation and decoding calls; performing tasks concurrently is not a reason to multiply their individual costs. Repeat that trajectory S times when independent sampling is required. Hoeffding inequality for range length $2B$ gives failure probability at most $2\exp\big(-S\varepsilon^2/(2B^2)\big)$, and rearrangement gives S. A sparse Hamiltonian multiply may cost its nonzero count instead. For generic n −qubit pure states $N = 2^n$, so polynomial dependence on N is exponential in n. A fixed-Hamming-weight sector has $N = \binom{n}{w}$, provided all relevant operations preserve it.

Scope. The first-248 encoding is not a Hamming sector. Arbitrary non-Clifford noisy physical-state simulation is not covered by a logical-register cost estimate or stabilizer tracking alone. Hamiltonian construction, optimization iterations, finite precision, output enumeration and measurement decomposition add costs. No efficient full tensor decomposition or 144-qubit simulation is proved.

Computational check. Exact binomial sampling tails versus Hoeffding, dimension and cost formulas, and a dense state-vector expectation example.

Unified validation and coverage

The merged ZIP contains original/ and update/ suites, with a single run_all.py entry point. It executes 32 original-suite checks and 18 updated-suite checks, including deliberately overlapping dependencies. These 50 checks are not 50 distinct theorems. Each suite verifies its own corrected text against this

consolidated document; a merged_manifest.json additionally verifies the restored introduction and consolidated bibliography.

Run python run_all.py --document merged_paper.docx --output merged_results.json after installing requirements.txt. Individual scripts remain available in each subfolder. PASS denotes successful execution of finite diagnostics, not formal proof of all statements or hardware certification. Original-version counterexamples are explicitly labeled. The optional NetworkX comparison reports NOT_RUN when unavailable; its exact small-graph oracle remains active.

Consolidated bibliography

Reference labels R1 through R6 used in Part II are retained. Remaining references from both papers appear as B1 through B13. Duplicate entries are listed once. Bibliographic retention does not validate claims attributed to a source in the uncorrected drafts.

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